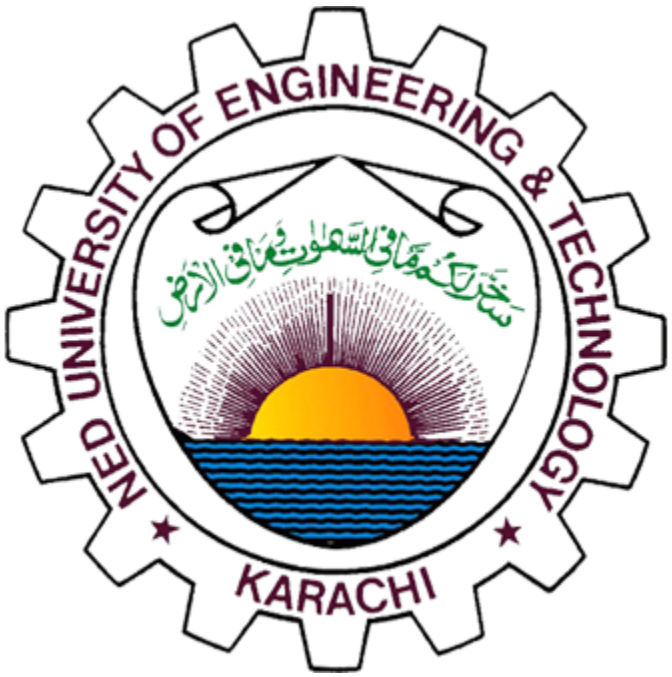

CS - 329
Operating System

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Fruit Picking Synchronization Problem with GTK GUI Visualization

1. Introduction

This project implements a classic producer-consumer synchronization problem (a model where some threads produce data and others consume it) using POSIX threads in C programming language. The system simulates a real-world fruit harvesting scenario involving multiple worker threads (pickers) and a loader thread responsible for transporting crates. Additionally, a GTK-based GUI is implemented to visualize thread activity, crate status, and system behavior in real time.

The goal is to ensure:

- Safe concurrent access to shared resources where multiple threads work without conflict
- Proper synchronization between threads (ensuring correct execution order)
- Fair workload distribution among pickers
- No race conditions, deadlocks (infinite waiting), or starvation (no execution chance)

2. Problem Description

The system consists of:

- **3 Picker Threads (Producers)**
  Pick fruits from a shared tree and place them into a crate.
- **1 Loader Thread (Consumer)**
  Waits until a crate is full, then loads it onto a truck and replaces it.

2.1 Constraints

- Each crate has a fixed capacity of 12 fruits
- Pickers can only place fruit when crate is OPEN (available)
- Loader acts only when crate becomes FULL (12/12 slots)
- Final partially filled crate must also be handled
- Fairness must be ensured among pickers

3. Synchronization Design

3.1 Shared Resources

- Tree (array of fruits)
- Crate (slots, state)
- Counters (next fruit index, crate count, etc.)

3.2 Synchronization Mechanisms

Mutexes (thread locks)

- tree_mutex → protects fruit picking
- crate_mutex → protects crate operations

Condition Variables (wait/notify mechanism)

- cond_crate_open → pickers wait if crate unavailable (pause)
- cond_crate_full → loader waits for full crate (until notified)

3.3 Crate State Machine

OPEN → FULL → SWAPPING → OPEN

State	Description
OPEN	Pickers can add fruit
FULL	Crate filled (12/12)
SWAPPING	Loader replacing crate

4. Algorithm Design

4.1 Picker Thread Logic

- Lock *tree_mutex* to pick fruit.
- Unlock and move to the crate.
- Lock *crate_mutex* and wait if not OPEN.
- Place fruit; if 12th fruit, signal loader.

4.2 Loader Thread Logic

- Lock *crate_mutex* and wait for FULL state or all pickers to finish.
- If full, mark the state as SWAPPING and load onto the truck.
- Replace with a new crate and broadcast *cond_crate_open* to all pickers.
- Handle any final partial crates before exiting.

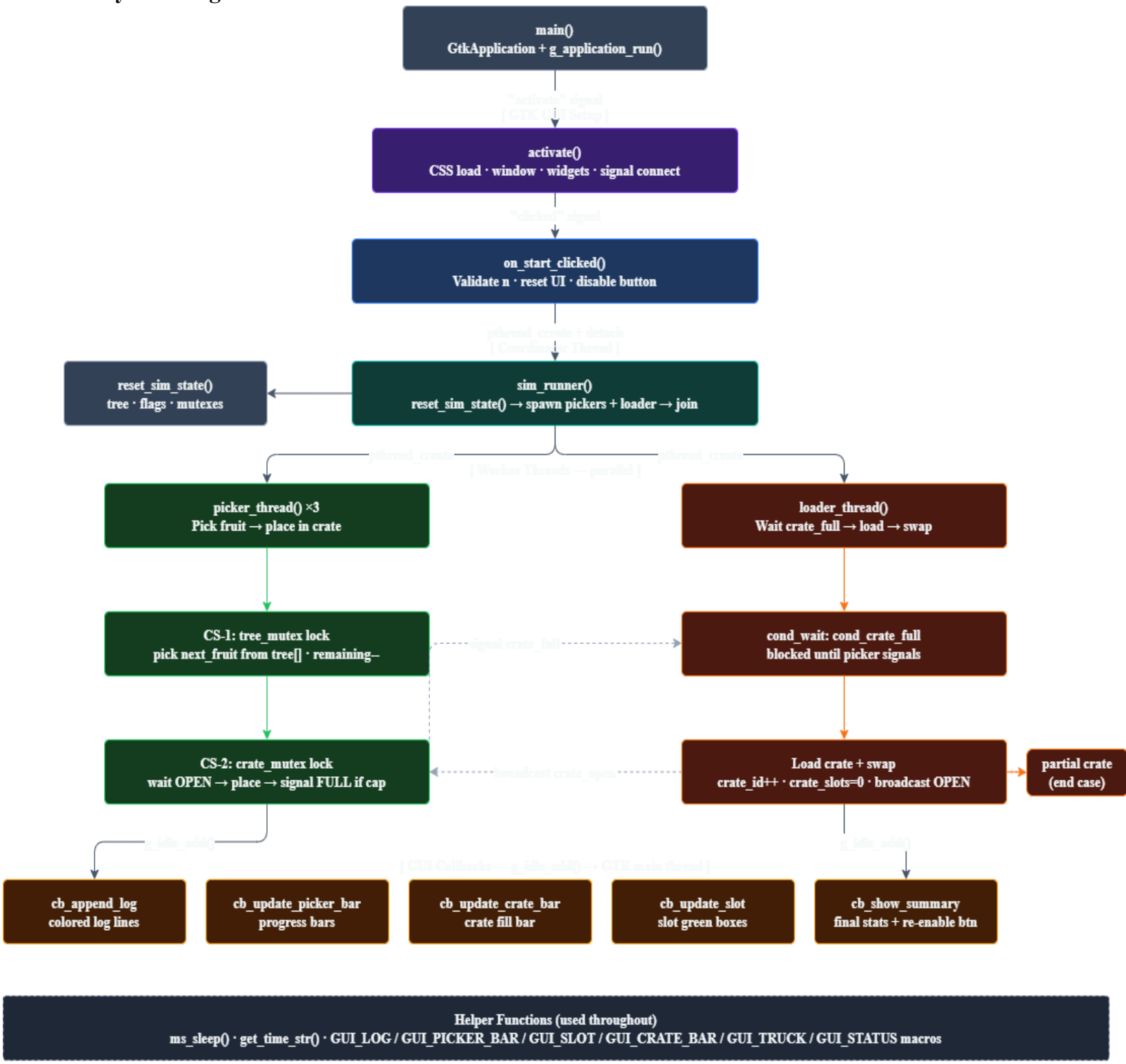
4.3 Thread Lifecycle

- Born → Spawned via *pthread_create()*.
- Running → Picking fruit or loading the crate.
- Blocked → Waiting on a mutex or *pthread_cond_wait()*.
- Signaled → Awakened via *pthread_cond_signal()* or *broadcast()*.
- Terminated → Exiting via *pthread_join()* after the tree is bare.

4.4 Inter-Thread Communication Summary

Source (From)	**Destination (To)**	**Signal Type**	**Reason / Trigger**
Picker	Loader	cond_signal	Crate full
Loader	All Pickers	cond_broadcast	New crate ready
Last Picker	Loader	cond_signal	Tree bare

4.5 System Logic Flowchart



5. Implementation Details

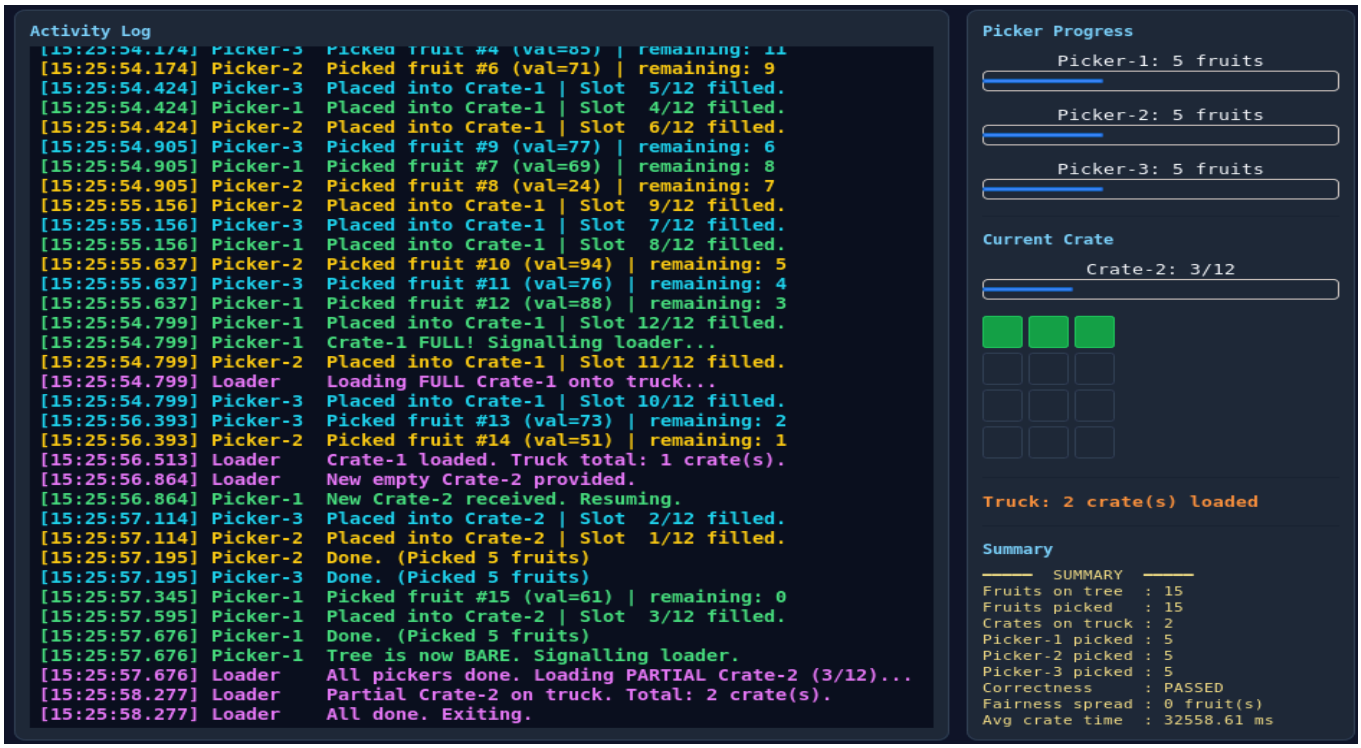
- **Language:** C
- **Standard:** POSIX
- **Compiler:** gcc
- **Thread Library:** pthread
- **GUI Library:** GTK+ 3 (used for visualization)
- **Signal Handling:** signal.h for SIGINT graceful exit
- **Timing Library:** clock_gettime(CLOCK_MONOTONIC) for performance metrics
- **GUI Thread Safety:** g_idle_add() ensures all widget updates happen on the main thread via heap-allocated callback structs, preventing GTK threading violations.
- **Compilation:** gcc -o Threads_GTK Threads_GTK.c -lpthread $(pkg-config --cflags --libs gtk+-3.0)
- **Execution:** ./Threads_GTK

6. Code

Link: https://drive.google.com/file/d/1hfDjVH4u2tzQiQga9v6ojDeebt070aA1/view?usp=sharing

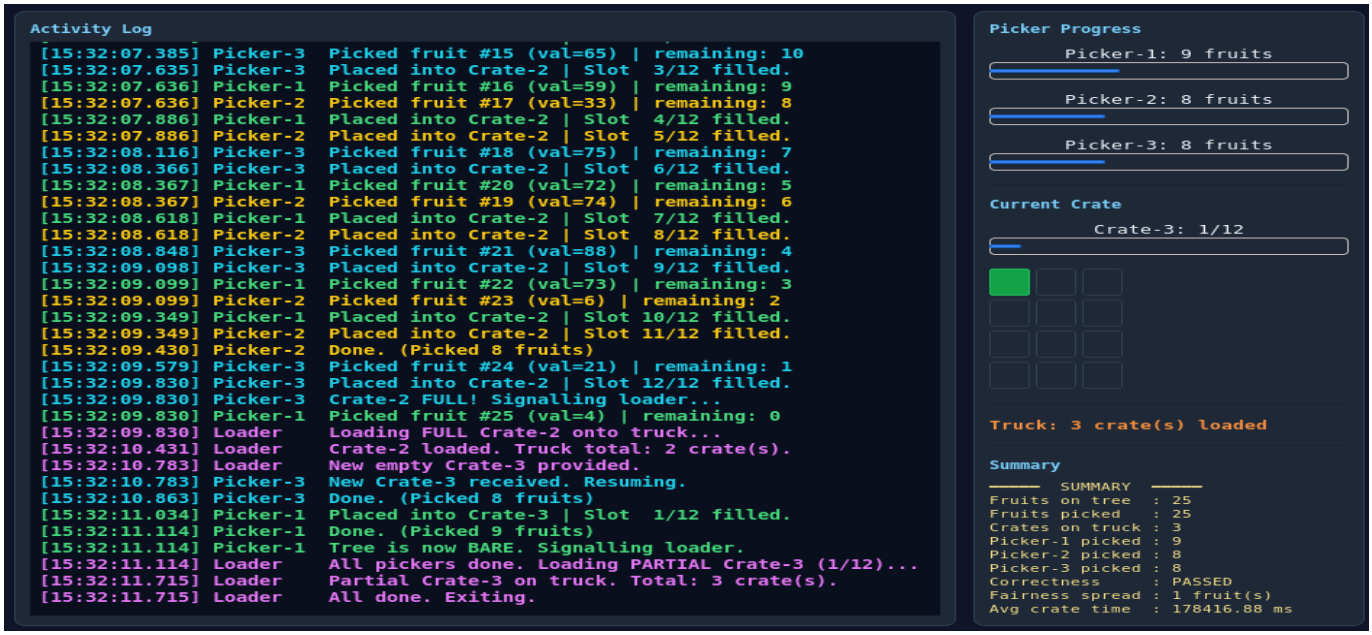
7. Test Cases

Test Case 1
Input: Number of fruits = 15
Expected Behavior:
→ Crates Loaded:1 full (12) + 1 partial (3 fruits)
→ Loader handles partial crate at end
→ Equal distribution among pickers
Output Snapshot:

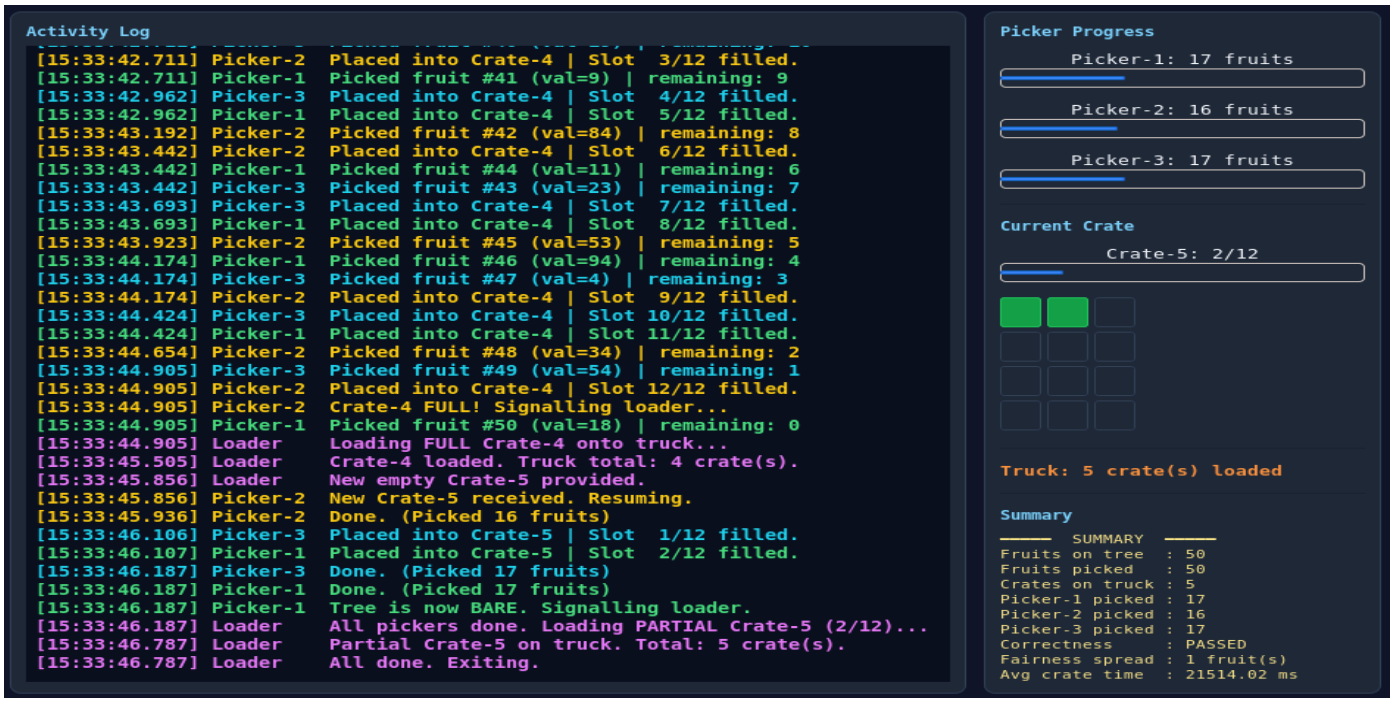


Test Case 2
Input: Number of fruits = 25
Expected Behavior:
→ Crates Loaded: 2 full (24) + 1 partial (1 fruit)
→ Loader handles partial crate at end

Output Snapshot:

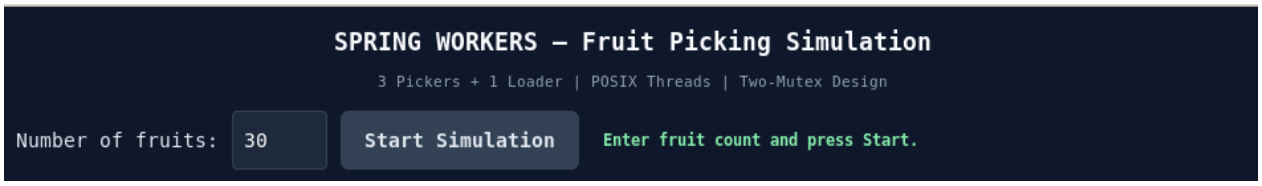


Test Case 3
Input: Number of fruits = 50
Expected Behavior:
→ Multiple crates loaded 4 full (48) + 1 partial (2 fruits)
→ Loader handles partial crate at end
Output Snapshot:

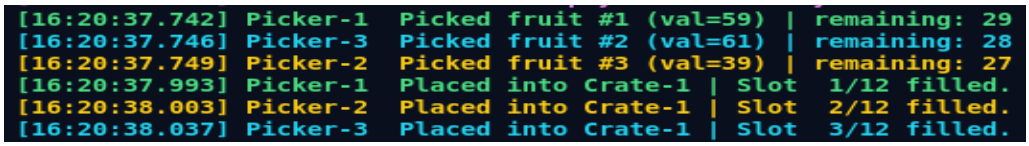


8. Execution Snapshots

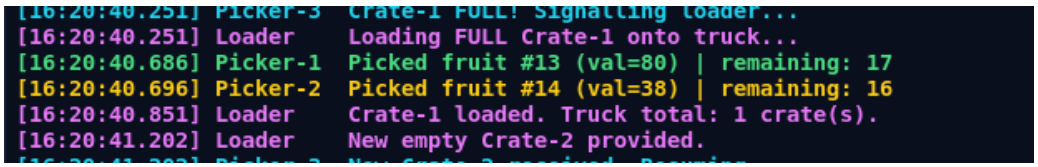
Program Start



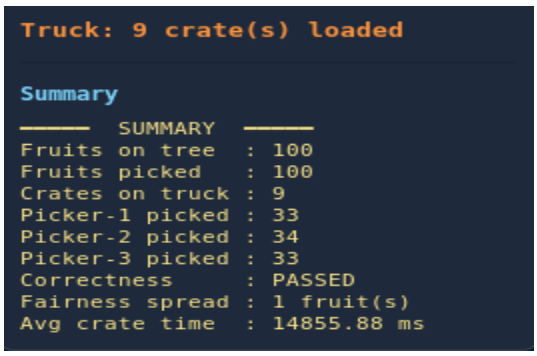
Picker Activity

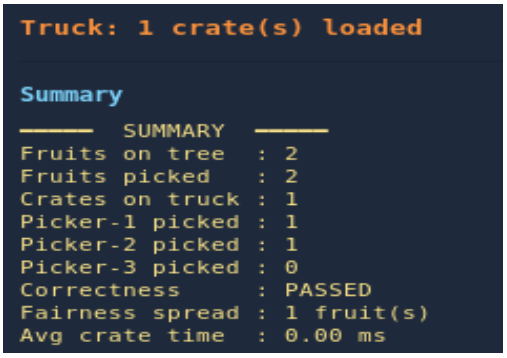


Loader Activity



Final Summary Output





9. Complexity Analysis

9.1 Mutex and Condition Variable Usage

- **tree_mutex:** Protects the fruit count. It is held only for a split second to increment the counter.
- **crate_mutex:** Protects the crate slots. It ensures only one picker writes to a slot at a time.
- **Design Rule:** Both mutexes are *never* held simultaneously. This simple rule completely prevents deadlocks. Two separate condition variables handle signals between pickers and the loader without confusion.
- **cond_crate_open**: Pickers wait on this variable when *crate_state != CRATE_OPEN*. The picker sleeps here releasing *crate_mutex* automatically until the loader calls *pthread_cond_broadcast(&cond_crate_open)*. This ensures no CPU cycles are wasted during the wait.
- **cond_crate_full**: The loader waits on this variable until either a picker signals that the crate is full via *pthread_cond_signal(&cond_crate_full)*, or the final picker signals that the tree is bare via *pickers_done == NUM_PICKERS*.
- **Separation of Concerns**: This two-condition-variable design cleanly separates communication channels pickers notify the loader through *cond_crate_full*, while the loader notifies pickers through *cond_crate_open*.

9.2 Race Condition Prevention

- **Double-picking:** The mutex prevents two pickers from grabbing the same fruit index.
- **Slot Overlap:** The mutex ensures no two pickers try to put a fruit into the same crate slot.
- **Early Loading:** The loader waits for the *CRATE_FULL* status, ensuring it doesn't empty the crate while a picker is still placing a fruit.
- **Completion Handshake:** The loader's use of *pickers_done* counter ensures that no fruit is left in limbo when the tree becomes bare, only after all pickers have finished does the loader process the final partial crate.
- **Performance Tracking:** The system tracks average crate filling time using *clock_gettime(CLOCK_MONOTONIC)*. Each time the loader provides a new crate, a timer starts. When the 12th slot is filled, elapsed time is recorded in *total_crate_time* and averaged across all crates. This metric appears in the final GUI summary as "*Avg crate time*" in milliseconds, helping identify picker throughput.
- **Slot Overflow Prevention:** The *CRATE_FULL* state is set inside *crate_mutex* before unlocking, ensuring no other picker can enter the crate between the state change and the unlock. This closes a subtle race window that would otherwise allow slot overflow.

9.3 Deadlock & Starvation Analysis

- **Mutual Exclusion:** Mutexes ensure only one thread accesses shared resources at a time.
- **Fixed Lock Order:** *tree_mutex* is always acquired before *crate_mutex*, eliminating circular wait and deadlock risk.
- **Separate Critical Sections:** tree_mutex is fully released before *crate_mutex* is acquired so no thread holds both simultaneously, breaking the hold-and-wait condition**.**
- **Auto-Release on Wait:** *pthread_cond_wait* atomically releases the mutex while sleeping, keeping other threads unblocked.
- **Final Broadcast:** At simulation end, a broadcast wakes any remaining waiting threads to ensure clean exit.

9.4 Edge Cases Handled

- **Perfect Multiples:** If the fruit count is a multiple of 12, the loader exits cleanly without trying to process an empty 13th crate.
- **Empty Tree Mid-Task:** If the tree runs out of fruit while a picker is working, they are allowed to finish placing their last fruit before exiting.
- **Fault Tolerance:** *A SIGINT* handler registered via signal(*SIGINT, handle_sigint*) ensures that if the simulation is force-closed with *Ctrl+C*, both mutexes are unlocked, all waiting threads are broadcast, and tree memory is freed before exit preventing resource leaks and deadlocks on abnormal termination.

9.6 Limitations

- **Bottleneck:** Since there is only one crate, pickers must wait for each other, which slows down the process.
- **Single Failure Point:** If the loader thread crashes or hangs, the entire system stops.
- **Memory:** Very large fruit counts could crash the program because there is no limit on memory allocation (malloc).
- **Compatibility:** The code is designed for Linux (POSIX/GTK) and will not run on Windows without major changes.

10. Conclusion

This project successfully implements a multi-threaded fruit harvesting simulation using POSIX threads, mutexes, and condition variables in C. Three picker threads and one loader thread run in parallel, safely sharing a tree and crate without race conditions, deadlocks, or starvation. Beyond basic synchronization, the implementation includes real-time performance tracking via *clock_gettime*, graceful fault tolerance via *SIGINT* handling, and GUI thread safety via *g_idle_add* bringing it closer to a concurrent system design that reflects real-world operating system challenges.

11. Future Enhancements

- **Runtime Configuration:** Make *NUM_PICKERS* and *CRATE_CAPACITY* adjustable via GUI sliders, no recompilation needed.
- **Parallel Crates:** Give each picker its own buffer crate to eliminate the placement bottleneck, only the loader touches the shared truck-bound crate.
- **Timeout on Waits:** Replace *pthread_cond_wait* with *pthread_cond_timedwait* so pickers exit gracefully if the loader hangs instead of freezing forever.
- **Speed Control:** Add a real-time slider to adjust *DELAY_PICK_MS*, *DELAY_PLACE_MS*, and *DELAY_LOAD_MS* for better observation of thread behavior.
- **Warehouse Scaling:** Support N pickers and M parallel crates via a lock-free queue, enabling warehouse-level simulations without restructuring core synchronization logic.
- **Priority Inversion Protection:** Initialize *crate_mutex* with *PTHREAD_PRIO_INHERIT* via *pthread_mutexattr_set* protocol to prevent a high-priority loader from being delayed by a lower-priority lock-holding picker.